James Zhang

Senior Full-Stack AI & automation Engineer

Bachelor of Science in Computer Science

Tsinghua University, Beijing, China

Graduated: 2015

- Focused on Artificial Intelligence, Machine Learning, and Software Engineering.
- Completed coursework in Data Structures, Algorithms, Computer Vision, and Deep Learning.
- Conducted a project on image processing using OpenCV and Python, applying machine learning techniques for object detection.
- Engaged in self-driven learning of web technologies including MERN stack and cloud deployment.

FAW Group Co Ltd Changchun, China

- **Senior Full-Stack AI & Automation Engineer**

July 2023 - PRESENT

- Delivered RAG + LLM assistants for engineering docs (chunking + vector database retrieval using LLM APIs), so teams get faster, more accurate answers
- Built Python AI pipelines for computer-vision defect detection using OpenCV + TensorFlow, improving model quality via data curation and tuning
- Implemented embedding + retrieval workflows (vector DB indexing, relevance filtering) to keep generated responses accurate as content updates
- Automated data processing and analysis using Python + SQL, orchestrated with `n8n`/`Make.com`/`Zapier`, with repeatable outputs for Excel and Power BI reporting
- Built KPI dashboards and automated report generation with Power BI + Excel, including VBA macros to reduce manual formatting and consistency issues
- Shipped React/JavaScript front-end and Node.js back-end APIs that turn ML outputs into clear KPIs and operational insights
- Created scheduled web-scraping and ingestion pipelines in Python using BeautifulSoup/Selenium to keep datasets current for training and reporting
- Coordinated and mentored a team of 4 engineers (reviews, planning, CI/CD reliability) to deliver dependable production services

- **Full-Stack Developer (AI/Computer Vision)**

April 2018 - June 2023

- Delivered RAG + LLM assistants for engineering docs (chunking + vector database retrieval using LLM APIs), so teams get faster, more accurate answers
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- Junior Developer / Automation Engineer

September 2015 - March 2018

- Automated data extraction and preprocessing using Python with BeautifulSoup/Selenium, reducing manual collection effort for recurring business tasks
- Developed and maintained internal PHP + MySQL applications for task management and operational workflows
- Built ETL and data-cleaning routines in Python to standardize datasets for analysis and model evaluation
- Supported report automation and data analysis by preparing consistent outputs for Excel/BI-style consumption and validating data quality before use

- Software Engineering Intern

July 2013 - August 2015

- Assisted with ETL and data cleaning routines in Python to standardize incoming datasets for downstream processing
- Helped prototype lightweight services and scripts to transform, validate, and format data for internal tooling needs
- Contributed to front-end work with JavaScript improvements and bug fixes for internal web pages

- Supported automation tasks by documenting workflows and helping verify job outputs to ensure data pipelines ran reliably

- **AI Development**: RAG, AI Agent, AI Chatbot, LLM
- **Automation Tools:** n8n, Make.com, Zapier
- **Frameworks & Libraries:** FastAPI, Flask, MERN, Django, ASP.NET, Laravel, WordPress
- **Databases:** MySQL, PostgreSQL, SQLite, MongoDB, SQL Server, Pinecone, Qdrant
- **Tools & Platforms:** Git, Docker, AWS, Cursor, VS Code, Visual Studio
- **AI & Machine Learning Libraries:** TensorFlow, Keras, scikit-learn, OpenCV, PyTorch
- **Web Scraping:** BeautifulSoup, Scrapy, Selenium
- **Image Processing:** OpenCV, PIL/Pillow, OCR
- **Data Processing & Automation:** Data cleaning, ETL pipelines, scripting for task automation, workflow orchestration
- **Bot Development:** Building chatbots, automation bots, integration with messaging platforms (e.g., Telegram, Messenger)
- **Office Tools:** Excel, Google Docs, Google Sheet, Google Form
- **Languages:** Python, JavaScript (Node.js, React, Angular), PHP, VB.NET, C#/C++/C

- **Chinese:** Native or Bilingual Proficiency
- **English:** Full Professional Proficiency